Worksheet 8

due at 23:59 November 9th (to be discussed November 10th)

Please include your full name and student ID directly on your homework submission.

1 Phase Kickback

Let U be a unitary operator with eigenstate $|\psi\rangle$ and eigenvalue $e^{2\pi i\theta}$:

$$U|\psi\rangle = e^{2\pi i\theta}|\psi\rangle.$$

(a) Consider a two-qubit system initialized in state $|+\rangle|\psi\rangle$, where $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$. Suppose we apply the gate U to the qubit in state $|\psi\rangle$ conditioned on the first qubit being in state $|1\rangle$. What is the resulting two-qubit state?

(b) Let $U = Z$ and $|\psi\rangle = |1\rangle$ (with $Z|1\rangle = -|1\rangle$). What is the state of the control qubit after the controlled-Z gate?

2 Binary Fractions

In a QPE circuit with m ancilla qubits, the jth ancilla qubit controls the application of U^{2^j} for $j = 0, \ldots, m-1$. Assume the phase θ, defined as above, can be expressed exactly as an m-bit binary fraction:

$$\theta = 0.\theta_1\theta_2\cdots\theta_m = \sum_{k=1}^{m} \theta_k 2^{-k}, \quad \theta_k \in \{0,1\}.$$

(a) What is the decimal representation of the binary fraction 0.111? What is the binary representation of the decimal fraction 0.625? Hint: Note that $0.625 = 5/8$.

(b) Show that for any integer $j \geq 0$, the phase produced by U^{2^j} acting on $|\psi\rangle$ satisfies

$$e^{2\pi i(2^j\theta)} = e^{2\pi i(0.\theta_{j+1}\cdots\theta_m)}.$$

Hint: Try separating $2^j\theta$ into integer and fractional parts, and don't forget that $e^{2\pi im} = 1$ for any $m \in \mathbb{Z}$.

(c) Suppose $\theta = 0.625$. Use your results from (a) and (b) to determine the phase factors applied by U^{2^0}, U^{2^1}, and U^{2^2}.

3 The Quantum Fourier Transform

Following the application of the controlled-U gates, the state $|\Psi\rangle$ of all qubits *prior* to the inverse Fourier transform $\mathrm{QFT}^\dagger$ is given by

$$|\Psi\rangle = |\psi\rangle \otimes \frac{1}{\sqrt{2^m}} \sum_{k=0}^{2^m-1} e^{2\pi i k\theta} |k\rangle .$$

In the case where θ is exactly equal to an m-bit binary fraction, we have $\theta = j/2^m$. The integer j can then be recovered as

$$|\psi\rangle \otimes |j\rangle = (\mathbb{1} \otimes \mathrm{QFT}^\dagger) |\Psi\rangle .$$

(a) Consider the case where $\theta \neq j/2^m$. Write the action of $(\mathbb{1} \otimes \mathrm{QFT}^\dagger)$ on $|\Psi\rangle$. This can be left as a sum for now.

(b) Using your result from (a), explain why a measurement of the ancilla register now leads (for $\theta \neq j/2^m$) to a probability distribution over computational basis states. Hint: Express $\mathrm{QFT}^\dagger |\Psi\rangle$ so that the coefficients of the computational basis states $\{|j'\rangle\}$ take the form $\sum_{k=0}^{M-1} r^k$ for some r and M, and then apply the following result for summing geometric series:

$$\sum_{k=0}^{M-1} r^k = \frac{1-r^M}{1-r}.$$

4 Single-ancilla QPE

Standard QPE assumes we have m ancilla qubits, all of which are measured at the end of the circuit. In this exercise, we consider an alternative setup where we instead have a single ancilla qubit. This ancilla qubit will be measured m times, with a reset in between measurements, and will return a binary digit θ_k of the unknown phase at each measurement.

(a) Assume $U|\psi\rangle = e^{2\pi i\theta}|\psi\rangle$, where $\theta = 0.\theta_1\theta_2\cdots\theta_m$ is an exact m-bit binary fraction. In the first step ($k = m$), the ancilla is initialized to $|+\rangle$, and a controlled $U^{2^{m-1}}$ gate is applied to the target in state $|\psi\rangle$. Write the joint state of the ancilla and target qubits following this gate.

(b) Starting from your result for (a), consider a measurement of the ancilla qubit in the X basis. (Equivalently, you can think about first applying a Hadamard and then measuring in the Z basis.) Show that the measurement outcome $x_m \in \{0, 1\}$ deterministically recovers the least significant bit θ_m of the unknown phase. Hint: This will use the result you proved in Question 2(b).

(c) We now consider the following steps $k = m-1, m-2, \ldots, 1$. In each step, assume that the bits $\theta_{k+1}, \cdots, \theta_m$ are known, and that the ancilla has been re-initialized to $|+\rangle$ following the measurement in the preceding step. For step k, the gate applied is a controlled $U^{2^{k-1}}$ gate. We can deterministically extract θ_k from an X-basis measurement of the ancilla, provided we first apply a phase rotation gate $R_z(-\phi_k)$ to the ancilla. Show that the required rotation angle is given by

$$\phi_k = 2\pi \sum_{\ell=k+1}^{m} \theta_\ell 2^{-(\ell-k+1)}.$$